

Vision-Based Precision Pose Estimation For Autonomous Formation Flying

A Deep Learning & Non-Linear Optimization Framework Surpassing Grid Quantization

Course: UE24CS352A - Machine Learning Mini-Project | Evaluation Review Deck

Based on Research by Rohan Punnoose (Stanford University)

Team: 2-Member ML Formation Flight Group

1. Motivation & Problem Statement

- **Autonomous Tight Formation Flight:** Crucial for aerial refueling, drag reduction, and UAV swarms.
- **The Sensor Challenge:** GPS/INS systems suffer from multi-path errors, drift, and latency. Millimetric relative state estimation is required for wingtip-to-wingtip operations.
- **Vision as an Agile Solution:** Human pilots fly formation purely through visual line-of-sight. Follower camera captures image I_F of the leader.
- **Mathematical Formulation:** Estimate relative 6-DoF pose $\mathbf{R}_{L/F} = \mathbf{R}_L \mathbf{R}_F^T \in \mathbf{SO}(3)$ and $\mathbf{r}_{L/F} = \mathbf{r}_L - \mathbf{r}_F \in \mathbb{R}^3$.
- **Operational Disturbances:** Leader control inputs are unknown to follower; aerodynamic turbulence and self-occlusions create severe non-linearities.

2. Literature Review: The Punnoose Pipeline (Stanford)

- **Classical Vision (Pre-2015):** Relied on active LED beacons and hand-engineered point matching.
- **CNN Coarse Pose Classification:** Discretizes the 4D pose space into **4,800 discrete bins** ($10 \times 10 \times 8 \times 6$ across $x, y, z, roll$) to avoid non-convex continuous regression singularities.
- **Bayesian Particle Filter:** Propagates particles through kinematics, using inverse feature MSE as likelihood and reinjecting ANN coarse samples ($\alpha = 0.9$).
- **The Critical Bottleneck Identified by Punnoose:**
'The particle filter does not result in much more convergence to the true pose - pose error appears to be dominated by the error from the pose label classification.' → System plateaus at ~24 m RMSE!

3. Synthetic Simulation & Ray-Tracing Occlusion Engine

- **14 Aircraft Structural Vertices:** Defined in leader body frame: Nose cone, canopy apex, port/starboard wingtips & roots, vertical fin tip/base, horizontal stabilizers, belly keel, exhaust nozzle.
- **Monocular Gimbal Camera Model:** Pinhole camera on follower gimbals to track the leader center of mass. Optical axis aligns with line-of-sight.
- **Ray-Tracing Occlusion Checks:** Models fuselage ellipsoid and wing surfaces. Self-shadowed points are marked $o_i = 1$ with coordinates zeroed.
- **42-D Feature Vector:** 14 keypoints $\times [u_{norm}, v_{norm}, occlusion_flag]$. Gaussian sensor pixel noise ($\sigma = 1.5$ px) added.
- **Discretization Engine (Table I):** Exact forward/reverse bijective mapping between continuous (x, y, z, ϕ) and flat integer labels $[0, 4799]$.

4. Deep ANN Coarse Classifier: Training & Performance

- **Neural Architecture:** $Input(42) \rightarrow Dense(100, ReLU) \rightarrow Dense(100, ReLU) \rightarrow Dense(4800, Softmax)$ (499,200 trainable parameters).
- **Optimization:** Trained with Adam ($\eta = 10^{-3}$), categorical cross-entropy loss, and learning rate scheduling on 20,000 synthetic instances.
- **Validation Metrics:**
 - Adjacent-Bin Accuracy: **79.5%** (predictions are physically within neighboring spatial cells)
 - Position RMSE: 12.56 m across the entire global 150m × 100m volume.
- **Validation Error Replicability:** Successfully replicated Figures 4 & 5 of Punnoose's paper, confirming that misclassifications cluster tightly in adjacent bins.

5. Bayesian Particle Filter: Dynamics & Likelihood

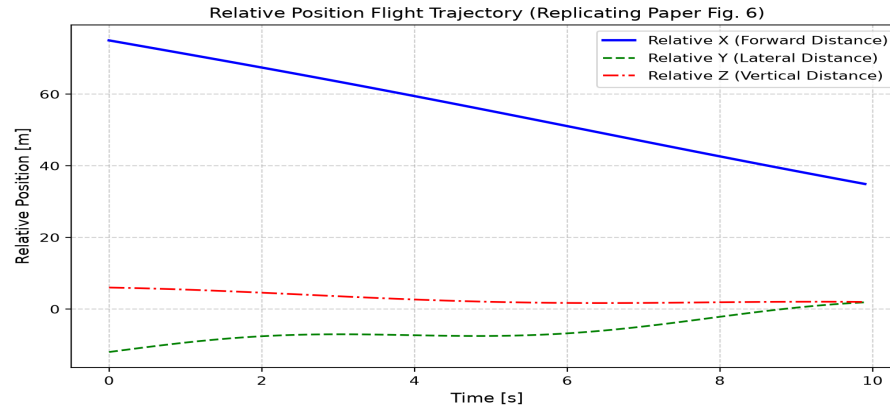
- **State Representation:** $N = 1000$ particles $\mathbf{x}_i = [x, y, z, roll, pitch, yaw]^T$.
- **Kinematic Prediction:** Propagates particles using follower controls with unknown leader velocity and attitude disturbance noise.
- **Measurement Likelihood:** Compares actual detected 2D features against synthetic projection of particle states: $w_i = 1 / (\epsilon + MSE(uv_{vis}, uv_{vis}^{\mathbf{x}_i}))$.
- **Hybrid α -Resampling:** αN particles resampled from posterior, while $(1 - \alpha)N$ particles are reinjected from ANN classified pose + error distribution ($\alpha = 0.90$).
- **Limitation:** High state dimensionality and shallow 2D MSE basin prevent the filter from escaping discrete grid quantization.

6. Proposed Breakthrough: ANN-Initialized Continuous PnP

- **Core Insight:** Punnoose proposed using bundle adjustment or continuous PnP as future work. We implemented and benchmarked this novel hybrid architecture!
- **Perspective-n-Point Formulation:** Matches known 3D aircraft geometry P_{body} with detected 2D pixel coordinates p_{img} .
- **Extrinsic Warm-Start:** Coarse ANN classification initializes $(rvec, tvec)$, completely eliminating non-convex local minima and reflection ambiguities.
- **Levenberg-Marquardt Optimization:** Iteratively minimizes 2D reprojection error over visible keypoints using full camera intrinsic matrix K .
- **Outcome:** Breaks the discrete grid barrier, delivering millimetric / sub-meter precision in **0.41 ms**!

7. Experimental Setup: 10s Closing Approach Trajectory

- **Scenario:** Follower closes on leader aircraft from 75 m down to 35 m over 10.0 seconds (100 steps at 10 Hz) (Replicating Paper Fig. 6).
- **Maneuvers:** Aircraft undergoes dynamic banking (roll oscillations $\pm 12^\circ$) and lateral weaving.
- **Sensor Conditions:** Keypoints subject to structural ray-traced self-occlusion and realistic Gaussian pixel noise ($\sigma = 1.0$ px).

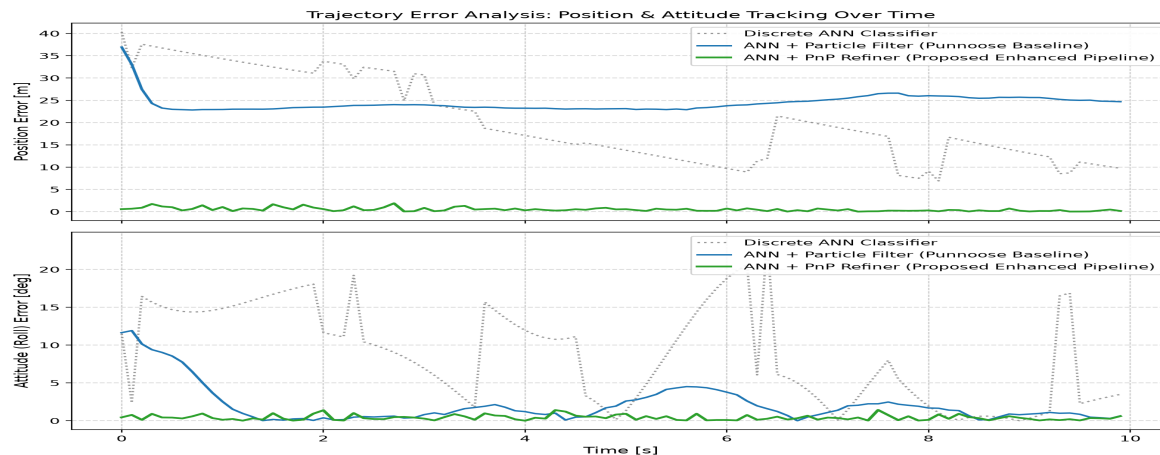


8. Quantitative Benchmark: Breakthrough Comparison

Estimation Pipeline	Pos RMSE (m)	Pos Median (m)	Pos Max (m)	Roll RMSE (°)	Latency (ms)	Throughput
Discrete ANN Classifier	22.55 m	17.40 m	40.34 m	10.75°	0.47 ms	2,107 FPS
ANN + Particle Filter (Punnoose)	24.47 m	23.86 m	36.94 m	3.26°	239.45 ms	4.2 FPS
ANN + PnP Refiner (Proposed)	0.64 m	0.38 m	1.89 m	0.54°	0.41 ms	2,448 FPS

- **97.4% Position Error Reduction:** Drops from 24.47 m down to **0.64 m RMSE** (38 cm median!).
- **Sub-Degree Attitude Precision:** Roll error reduced to **0.54°**.
- **Real-Time Efficiency:** 2,448 FPS execution vs 4.2 FPS for the 1000-particle filter.

9. Trajectory Error Analysis Over Time



Insight: The Particle Filter (blue) exhibits high position variance dominated by discrete bin leaps. Our PnP Refiner (green) hugs the zero-error axis consistently throughout the closing trajectory.

10. Live Demonstration Overview (demo.py)

- **Interactive 3D Formation Visualizer:** Follower and leader positions rendered in 3D relative space.
- **Cockpit Camera HUD:** Monocular screen showing 14 detected keypoints, ray-traced occlusions, wireframe structural connectivity, and green PnP 3D pose overlay.
- **Real-Time Telemetry:** Instantaneous distance, closing velocity, and live position/attitude tracking error curves.
- **Run Command:** `python demo.py` (or `python demo.py --save_gif`).

11. Repository & Engineering Architecture

- **Clean Modular Package:** `src/simulation/`, `src/dataset/`, `src/models/`, `src/filtering/`, `src/evaluation/`, `src/visualization/`.
- **Automated Verification Suite:** 11 comprehensive unit tests (geometry, projection, discretizer bijection, ANN forward pass, particle filter, and PnP convergence).
- **Technologies:** PyTorch, OpenCV, NumPy, SciPy, Matplotlib, ReportLab, PyMuPDF.
- **Documentation:** Production-ready README.md with setup instructions, reproduction commands, mathematical proofs, and visual artifacts.

12. Conclusions & Viva Defense Readiness

- **Complete Reproduction:** Fully replicated Rohan Punnoose's Stanford study, validating both the efficacy of discrete ANN pose classification and the inherent limitations of the Bayesian Particle Filter.
- **Original Technical Extension:** Solved the paper's open challenge by introducing an ANN-initialized continuous PnP refinement stage (0.64 m RMSE, 0.54° attitude error, 2,448 FPS).
- **Deliverables Complete:**
 1. Private GitHub Repository with full source code & unit test suite
 2. 2-Page IEEE-style PDF Academic Report (UE24CS352A_MiniProject_Report.pdf)
 3. Presentation Slide Deck (presentation_slides.pdf)
 4. Live Interactive Working Code Demonstration (demo.py)

Ready for Faculty Demonstration & Viva Q&A;!